Bus Tracker Hardware Specification

Oct 18, 2026 · @deeskhit

Document ID: HW-SPEC-1.0 | Status: Draft for review by the ECE team | Companion: SW-MVP-1.0, the software specification.

This document defines the tracker fitted to every bus: its parts, how they connect, what each part is responsible for, what the firmware sends, and how the software responds. Where the software behaviour is described, the software specification is the authority.

0. How to read this document

- MUST: required for the MVP. SHOULD: required unless a reason is recorded. MAY: optional.
- CONFIRM: a decision the ECE team must make. Section 22 lists them all.
- Proposed: a starting value or pin assignment. Verify it against the datasheet of the exact part before use.
- Values are given in metric units and in the units the software expects.

1. Purpose and scope

1.1 Purpose

Each bus carries one tracker. The tracker reports the bus's position and health to the server. It lets the driver raise an SOS. It keeps working, and keeps storing data, when the mobile network is unavailable.

1.2 In scope

Hardware parts, electrical connections, power, antennas, enclosure and mounting, firmware responsibilities, the data the firmware sends, failure behaviour, provisioning, installation, commissioning, testing, firmware updates, and maintenance.

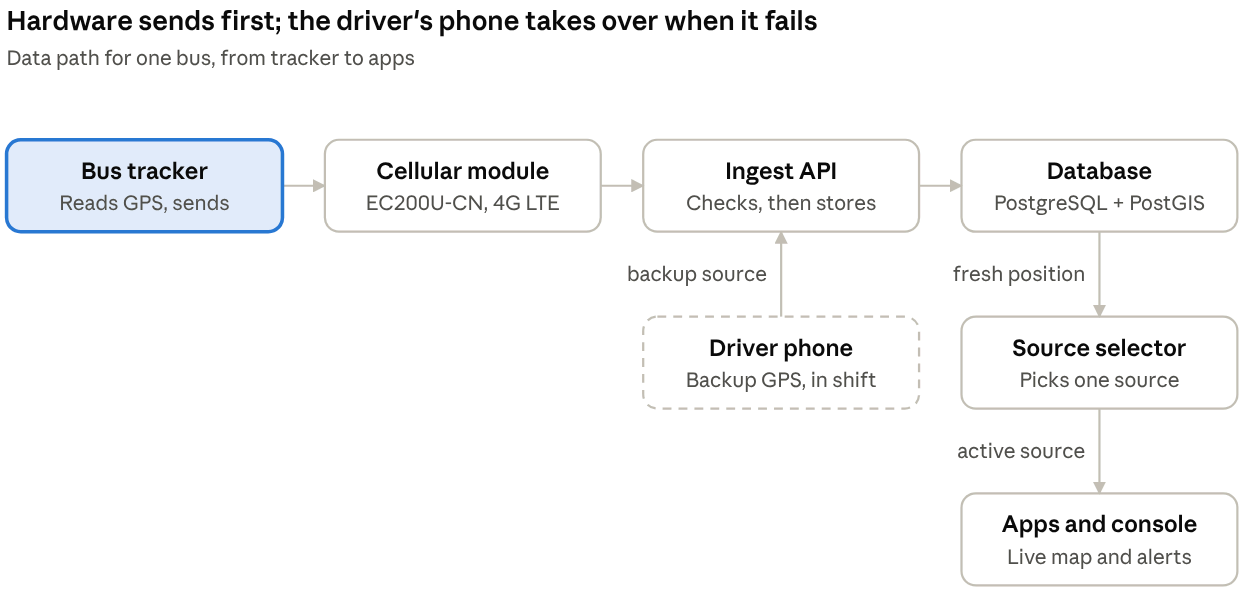
1.3 Out of scope

Server internals, the mobile and web applications, and the data model. These are in SW-MVP-1.0.

1.4 Fleet

Fifty or more buses, one tracker per bus. Each tracker has a unique device_id in the form HW_nnn. The coordinator links each device to one bus at a time. Devices are never linked to a bus by the technician or the installer.

2. System overview



data path for one bus: tracker to apps, with the driver phone as backup

Each bus has one tracker. The tracker sends its position through the cellular module to the ingest API, which checks and stores it. The driver's phone sends its position to the same API during a shift, and takes over as the source when the tracker stops sending.

The source selector decides which source is active for each bus. The apps read that result and show the source to the user. The hardware never decides what users see.

3. Responsibilities

Part or party	Responsible for	Not responsible for
Controller (ESP8266 proposed, ESP32 alternative)	Running the firmware: sampling, batching, buffering, sending, SOS handling, LED and buzzer, health reporting, applying server configuration	Deciding which source users see. Deciding which bus the device belongs to
Cellular module (Quectel EC200U-CN)	LTE connection, TLS, HTTPS requests made through AT commands, and GNSS if the order code includes it	Application logic, buffering policy, SOS decisions
GNSS receiver (inside the module, or a NEO-6M)	Satellite fixes, UTC time, speed, heading	Quality filtering. The firmware applies the quality rules
Power unit	Converting vehicle power to a stable supply, protecting against spikes and reverse polarity, charging the backup battery if fitted	Radio performance
Backup battery (if fitted)	Keeping the controller and module on for a short time after vehicle power is lost	Long-term operation
SOS button	Sending a physical press to the controller	Choosing the SOS category. The app and server do that
Ignition sense	Telling the controller whether the engine is running	Vehicle control of any kind
Status LED and buzzer	Showing device state to the driver and faculty, and confirming an SOS	Showing bus status to students. That is the apps' job
SIM (IoT or data plan)	Network identity and data	Routing or security decisions

Part or party	Responsible for	Not responsible for
Antennas, enclosure, wiring	Reception, protection, and reliable connection	Device logic
Server (software specification)	Validating, storing, choosing the active source, showing users, alerting, assigning buses, managing tokens	Physical device behaviour
Technician	Provisioning devices, signing off installations, replacing devices, approving firmware rollouts	Linking devices to buses
Coordinator	Linking devices to buses, revoking tokens, deciding rollout timing with the technician	Physical installation
Installer	Physical installation to section 14	Configuration in the server

4. Bill of materials per bus

Item	Quantity per bus	Purpose	Status
Controller board (ESP8266 or ESP32)	1	Firmware, buffering, watchdog, GPIO	CONFIRM. See section 5.3
Quectel EC200U-CN module on a Mini PCIe carrier, with SIM holder and antenna connectors	1	LTE, TLS, optional GNSS	Module chosen. Carrier board to select
GNSS receiver, NEO-6M class	0 or 1	Position, only if the EC200U-CN order code has no GNSS	CONFIRM

Item	Quantity per bus	Purpose	Status
LTE antenna, external	1	Cellular	Type and mounting CONFIRM
GNSS antenna, active with bias	1	Position	CONFIRM
SIM card, IoT or data plan	1	Connectivity	Plan and data cap CONFIRM
Power unit: automotive input, regulated output, protection	1	Power	Design in section 6
Fuse holder and fuse	1	Protection	Value in section 6
Backup battery pack with charger and protection	1	Keeps the SOS path alive after power loss	Recommended. CONFIRM
SOS button, red, momentary, guarded	1	Emergency	Required
Ignition sense interface, isolated	1	Ignition state	Required
Status LED, two colour or tri-colour	1	Device state	Required
Buzzer	1	SOS and fault feedback	Required
Enclosure, IP54 minimum	1	Protection	Rating CONFIRM
Wiring harness, connectors, cable ties, labels	1 set	Installation	Required
Door sensor, reed or switch	0 or 1	Door events	CONFIRM
Tamper switch	0 or 1	Tamper events	CONFIRM
RFID reader, MFRC522 class, with cards	0 in MVP, 1 in phase 2	Boarding by card	Phase 2

Spares: keep 5 percent of the fleet, and never fewer than 2 units. Spares are pre-provisioned by the technician (section 13) so they can be fitted quickly.

5. Electrical connections

The pin numbers come from the datasheets for the exact parts. The signal names and GPIO numbers here are proposals.

Logic levels: the EC200U-CN serial interface is expected to use 1.8 V logic, and the ESP8266 uses 3.3 V. Verify this in the EC200U-CN datasheet. Unless the datasheet confirms that the module's serial pins tolerate 3.3 V, put a level shifter between the module and the controller, in both directions.

5.1 Cellular module to controller

Signal	Module side	Controller side (proposed)	Notes
Serial data from module	Module TXD	Controller RX on UART0	Through a level shifter
Serial data to module	Module RXD	Controller TX on UART0	Through a level shifter
Ground	GND	GND	Short, low-impedance connection
Power on	PWRKEY	GPIO13 through an NPN transistor	Pulse length as the datasheet specifies
Hard reset	RESET_N	GPIO16 through an NPN transistor	Used only when the module stops responding
Module state	STATUS	Optional GPIO input	Used for diagnostics
Module supply	VBAT	Power unit output (section 6)	Within the module's supply range. Verify in the datasheet
LTE antenna	ANT (main)	LTE antenna through a short cable	50 ohm path

Signal	Module side	Controller side (proposed)	Notes
SIM	SIM socket	SIM card, IoT or data plan	APN set in firmware. CONFIRM

The controller's second serial port (UART1) is transmit-only on the ESP8266. It is used for debug logs in development only. Production units disable debug output (section 18).

5.2 GNSS

Option A (preferred, if the EC200U-CN order code includes GNSS):

- GNSS is run inside the module. There is no separate GNSS serial link.
- The GNSS antenna connects to the module's GNSS antenna connector.
- Position is read through the module's AT commands. The firmware reads fixes and time from the module.

Option B (if the order code has no GNSS):

- A NEO-6M class receiver connects to the controller: its TX goes to a receive-capable serial port on the controller. Its RX is optional, for configuration.
- The ESP8266 has only one receive-capable serial port, and the module needs it. So either use an ESP32, which has enough serial ports, or read the NEO-6M through a software serial port at 9600 baud and test it for dropped sentences on the bus. CONFIRM.
- NMEA speed (RMC sentence) is in knots. The firmware converts it to km/h (multiply by 1.852).
- The active antenna needs its bias supply from the receiver or a separate supply, as its datasheet specifies.

5.3 Controller pin map (proposed, ESP8266)

Function	Pin (proposed)	Direction	Electrical	Notes
Module serial	UART0 RX and TX	Both	3.3 V through level shifter	Disable console output on UART0 in production

Function	Pin (proposed)	Direction	Electrical	Notes
Debug log	UART1 TX (GPIO2)	Output	3.3 V	Development builds only
Module power on	GPIO13	Output	Through transistor	
Module reset	GPIO16	Output	Through transistor	No interrupt or pull-up on this pin
SOS button	GPIO4	Input	Pull-up, active low	Debounce in firmware
Ignition sense	GPIO5	Input	From isolated interface, active high	Debounce 500 ms
Status LED, red	GPIO12	Output	Through driver	
Status LED, green	GPIO14	Output	Through driver	
Buzzer	GPIO2 (shared with debug) or GPIO0	Output	Through transistor	GPIO0 and GPIO2 are boot-strapping pins. Check the boot states
Door sensor (optional)	Not assigned on ESP8266	Input		Needs a pin the ESP8266 does not have free. See below
Tamper switch (optional)	Not assigned on ESP8266	Input		Same

Pin shortage: the ESP8266 has too few free pins for every function in this list. The ECE team should either use an ESP32, which has enough pins and a second serial port, or add an I/O expander. The recommendation is the ESP32. CONFIRM.

5.4 Ignition sense

- The input comes from the ignition-switched circuit of the vehicle, at 12 V or 24 V nominal, according to the bus.

- It is electrically isolated from the controller, using an opto-isolator, with a series resistor sized for the full vehicle voltage range.
- A transient suppressor (TVS diode) protects it against vehicle electrical transients. Its rating is set for the vehicle class. CONFIRM the value with the vehicle electrical specification.
- IGNITION ON is reported when the input is above the ON threshold for 500 ms. The threshold is set in firmware and tested on the bench (section 16).
- Connect only to the ignition-switched circuit. Never connect it to the starter, lighting, or horn circuits.

5.5 SOS button

- A momentary push button, red, guarded or recessed, mounted within the driver's reach and not blocked by the steering wheel or the seat.
- Wired to GPIO4, with a pull-up to the controller's 3.3 V supply. Pressing the button pulls the line to ground. Cable length under 2 m. Twisted pair is recommended.
- Firmware debounce: a press is a low level sustained for 300 ms.
- Stuck-button rule: a press held longer than 10 seconds sends one SOS. The firmware then ignores the button until it is released.
- A cable break cannot be detected in the MVP. This is recorded as a known limit.

5.6 Status LED and buzzer

LED states are listed in section 9. The buzzer is used for SOS and for a startup failure. It does not sound continuously. Its loudness must be audible in the cabin with the engine running. CONFIRM the level by a bus test.

5.7 Programming and debug access

- A four-pin programming header or a USB serial adapter connects to the controller's serial pins.
- Access is behind the enclosure cover.
- Production firmware disables the serial console and never prints tokens or keys.

5.8 RFID reader (phase 2)

- An MFRC522 class reader over SPI: SDA (chip select), SCK, MOSI, MISO, RST, 3.3 V, and GND.

- Powered from 3.3 V only.
- Mounted at door step height, away from the engine. The reading distance is tested with the card type the college chooses.

5.9 Door sensor (optional)

A reed switch or micro switch on the door, with a 500 ms debounce. Needs a free pin (section 5.3).

5.10 Tamper switch (optional)

A normally closed micro switch on the enclosure lid. The switch opens when the lid is removed, and the firmware sends a TAMPER event. Needs a free pin (section 5.3).

6. Power

6.1 Requirements

- Input: vehicle battery, 12 V class or 24 V class. The fleet mix is CONFIRM. Each unit is configured for one class, and the class is recorded in provisioning.
- Protection on the input: a fuse at the battery side, reverse-polarity protection, and a transient suppressor rated for the load-dump transients of the vehicle class.
- A dedicated regulated supply for the cellular module, within the module's VBAT range, with capacity for its peak current. Do not power the module from the controller's 3.3 V rail.
- A separate 3.3 V regulator for the controller and sensors.
- Single-point ground connection, short and sized for the peak current.
- Fuse value: proposed 2 A for the whole device, CONFIRM after the bench measurement.

6.2 Load budget (estimates, to be measured on the bench)

Load	Typical average	Peak	Basis
Controller with radio active	About 80 mA at 3.3 V	About 300 mA	Estimate. Radio bursts raise the peak

Load	Typical average	Peak	Basis
Cellular module	Depends on network state	About 2 A during transmission	Peak to be confirmed in the EC200U-CN datasheet
GNSS receiver (NEO-6M class, if used)	About 45 mA	Not applicable	Estimate
LED and buzzer	Under 50 mA	About 80 mA while the buzzer sounds	Estimate
Ignition sense	Under 5 mA	Not applicable	Estimate

The measured totals decide the regulator rating and the fuse. Measure at the input with the module transmitting, in a worst-case signal condition.

6.3 Low-voltage cut-off

- Proposed cut-off: 11.0 V for the 12 V class, and 22.0 V for the 24 V class. CONFIRM against the vehicle electrical specification.
- Below the cut-off, the device stops transmitting, sends a LOW_VEHICLE_BATTERY event if it can, and enters the lowest-power state that still keeps the ignition sense active.
- Above the cut-off, normal operation resumes.

6.4 Backup battery (recommended)

- Purpose: keep the controller and module powered after the vehicle supply is lost, long enough to send an SOS and to flush the buffer.
- Minimum runtime: 10 minutes at idle, plus one SOS attempt. CONFIRM.
- Charged from the vehicle supply through a charger with current limit and temperature cut-off.
- A protection board prevents over-discharge, over-charge, and short circuit.
- Chemistry and enclosure are CONFIRM with the supplier. The pack must be mounted in a fire-resistant location, away from the engine.
- The device reports device_battery_mv in each heartbeat (software section 11.6).

6.5 Sleep behaviour

- Ignition ON: no sleep. The controller samples and sends at the cadence in section 10.
- Ignition OFF (parked): the controller uses its light-sleep mode between heartbeats. The module uses its low-power mode, using the AT commands in the EC200U-CN manual. Confirm that the module wakes reliably on the ignition interrupt before the design is frozen.

7. Antennas, enclosure, and mounting

- LTE antenna: external, mounted on the roof or upper dashboard, with the cable as short as the mounting allows. The maximum cable loss is set in the design. CONFIRM.
- LTE antenna routing: keep the cable at least 10 cm from high-current harnesses, wiper motors, and the engine bay.
- GNSS antenna: active, with a clear view of the sky. Mount it on the roof or on top of the dashboard. Avoid metallic or heavily tinted glass where possible. Keep it away from the LTE antenna.
- Enclosure location: under the dashboard, in an electrical box or ventilated bracket. Away from engine heat and out of the driver's leg space. Access is restricted to technicians.
- Operating temperature inside the bus: minus 10 to 55 degrees Celsius, proposed. Storage: minus 20 to 60 degrees Celsius. CONFIRM.
- Enclosure rating: IP54 minimum. Cable entries sealed with glands.
- Vibration: fixed with vibration-damping mounts, with torques set in the installation procedure (section 14).
- Labels: device ID (HW_nnn), a provisioning QR code, the fuse value, the bus voltage class, and the SOS button label.

8. Firmware responsibilities

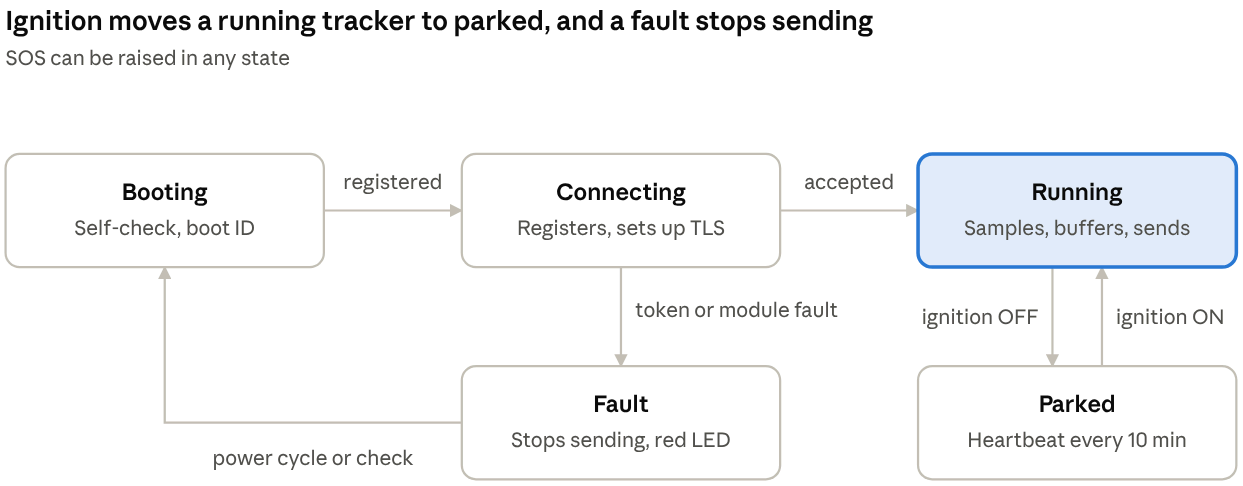
The firmware must do each of the following. The software behaviour that results from each item is in section 11.

1. Boot: run a self-check, read stored configuration, record the reset reason, and generate a new boot_id.

2. Connect: power the module, wait for network registration within a timeout, set the APN, activate the data context, and configure HTTPS with the certificate settings for the EC200U-CN.
3. GNSS: acquire fixes, apply quality rules (satellites of 4 or more, HDOP of 5.0 or less), set gps_valid, read time, and convert speed from knots to km/h when NEO-6M is used.
4. Sampling: take samples at the cadence in section 10.
5. Buffering: store each sample in a compact binary record (proposed 32 bytes) with its sequence number, in a flash partition of at least 512 KB. When the buffer is full, drop the oldest records and send BUFFER_OVERFLOW.
6. Sending: batch up to 50 samples per request. Send over HTTPS. Read the response, and delete only the samples that the response acknowledges (software section 11.4).
7. Retry: use the backoff and jitter in software section 11.4.
8. Events: detect and send the events listed in section 10.4.
9. SOS: handle the button as in section 5.5, and resend every 5 seconds until a response returns an incident_id.
10. Heartbeat: send every 60 seconds while ignition is ON, and every 10 minutes while parked. Include the fields in software section 11.6.
11. Configuration: apply the values returned in responses. Store them. Fall back to built-in defaults if stored values are missing or corrupt.
12. Watchdogs: a hardware watchdog on the controller. A software watchdog on the module: if the module does not answer AT commands for 30 seconds, restart the module. After three failed restarts, pulse the module's RESET line, and send MODULE_ERROR.
13. Time: use GNSS time when valid, otherwise module network time if available. Send CLOCK_SYNC when the clock changes.
14. Token: keep the device token in protected flash. Never log it. Rotate it as in software section 11.1.
15. Firmware update: act on pending_action from the heartbeat. Download the image, verify its SHA-256 and signature, write it to the inactive slot, reboot, and confirm. Roll back if the new image fails its checks (section 17).
16. Indicators: show state on the LED and buzzer as in section 9.
17. Production hardening: the serial console is disabled, and debug output is off (section 18).

9. Operating states

The tracker is always in one of five states. SOS can be raised from any state.



tracker states, with ignition and fault transitions

- Booting: self-check and read configuration. Moves to Connecting when the module registers, or to Fault after three module failures.
- Connecting: registers on the network, activates data, and makes its first request. Moves to Running when the server accepts a request. Moves to Fault if the token is rejected (401 or 403), or if the module fails repeatedly.
- Running: samples, buffers when the network is lost, and sends. Moves to Parked on ignition OFF. Network loss does not change the state.
- Parked: sends a heartbeat every 10 minutes. Moves to Running on ignition ON.
- Fault: stops uploading, shows a red LED, and checks the token every 15 minutes. Recovers after a power cycle or a passing check.

LED patterns:

LED	Meaning
Green, solid	Connected, valid GNSS fix, last heartbeat sent
Green, slow blink	Connected, no valid GNSS fix
Amber, solid	Booting or connecting
Amber, blink	Buffering, no connection

LED	Meaning
Red, solid	Fault, or SOS pending
Red, slow blink	SOS confirmed by the server, for 60 seconds
Off	Unpowered, or in the lowest-power state

10. Data the hardware sends

The message formats are defined in software specification sections 11 and 12, which are the authority. This section defines when the hardware sends each message, and where each value comes from.

10.1 Sending cadence

Condition	Samples	Batch send	Heartbeat
Ignition ON, moving (speed above 3 km/h)	Every 10 seconds	Every 10 seconds, or at 10 samples	Every 60 seconds
Ignition ON, stopped	Every 30 seconds	Every 30 seconds, or at 10 samples	Every 60 seconds
Ignition OFF (parked)	None	None	Every 10 minutes
SOS	None	Immediately	As usual
Buffer flush after a loss	No new samples	At most one request every 2 seconds	As usual

Without a valid fix, the firmware treats the bus as stopped.

10.2 Sample fields and sources

Field	Source	Units and rules
seq	Counter, reset on each boot	Increases by 1 per sample
boot_id	Random value at each boot	8 hex characters

Field	Source	Units and rules
recorded_at	GNSS time when gps_valid is true, otherwise null	UTC
gps_valid	Firmware quality check	True only when satellites is 4 or more and hdop is 5.0 or less
lat, lng	GNSS	Decimal degrees, 6 decimal places
speed_kmh	GNSS speed	km/h. Converted from knots when NEO-6M is used
heading_deg	GNSS course	0 to 360
satellites, hdop	GNSS	Count and dilution of precision
ignition	Ignition sense input	ON, OFF, or UNKNOWN
vehicle_battery_mv	Divider on the ADC	Millivolts. The ESP8266 analogue input accepts only about 1.0 V, so the divider ratio is set in the design

10.3 Heartbeat

The heartbeat fields are listed in software section 11.6. The module supplies network status, SIM status, and signal strength through its AT commands. The controller supplies uptime, buffer use, and the GNSS status.

10.4 Events sent by the hardware

The server's handling of each event is in software section 12. This table gives the hardware trigger and timing only.

Event	Hardware trigger	Timing and notes
BOOT	Every power-up or reset	Sent after the first successful registration. Carries the reset reason
IGNITION_ON	Ignition input goes ON	Debounced for 500 ms
IGNITION_OFF	Ignition input goes OFF	Debounced for 500 ms

Event	Hardware trigger	Timing and notes
SOS	SOS button held for 300 ms	Sent immediately. Resent every 5 seconds until acknowledged (section 5.5)
GPS_LOST	No valid fix for 30 seconds while ignition is ON	Sent once per loss
GPS_RECOVERED	First valid fix after GPS_LOST	Sent once per recovery
NETWORK_RESTORED	Registration returns after a loss	Sent first in the flush, followed by buffered samples
BUFFER_OVERFLOW	Oldest records dropped to make room	Sent with the dropped sequence range
LOW_VEHICLE_BATTERY	Vehicle voltage below the class threshold for 30 seconds	Sent once per episode
LOW_DEVICE_BATTERY	Backup battery below its threshold	Sent once per episode
MODULE_ERROR	Module error or unexpected reset	Sent after the module recovers
CONFIG_APPLIED	New configuration stored	Carries the configuration version
FIRMWARE_UPDATED	New firmware confirmed after a reboot	Carries the old and new versions
FIRMWARE_FAILED	Update failed and rolled back	Carries the attempted version
TAMPER	Tamper switch opens (only if fitted)	Immediate
DOOR_OPEN, DOOR_CLOSE	Door sensor changes (only if fitted)	Debounced for 500 ms
RFID_TAP	Card read (phase 2)	Carries the card UID, which the server hashes

Event	Hardware trigger	Timing and notes
CLOCK_SYNC	Clock corrected by GNSS or network time	Carries the old and new times

Events are kept in flash with the same care as samples. An event is removed only after the server acknowledges it.

11. What the software does with each hardware message

This table is the short form of software specification sections 11 to 13. Use it to check whether the firmware and the software agree.

Hardware message	Software action	Result users or staff see
Sample with gps_valid true, fresh	Stored. Becomes the hardware live state if it is the newest valid sample	Bus shows Live, bus tracker
Sample with gps_valid false	Stored. Not used for live state	After 60 seconds without a valid fix, the source may switch to the phone
Sample older than 10 minutes (buffered)	Stored as BUFFERED. History only	Visible in trip history, not in live view
Sample at 0, 0	Rejected with NULL_ISLAND	None
Sample implying a jump above 150 km/h	Stored and flagged. Rejected with IMPLAUSIBLE_JUMP	Not shown live
Duplicate sample (same boot, same seq)	Accepted without a new record	None
Same seq with different content	Rejected with CONFLICT	Technician sees the rejection count
Batch of up to 50 samples	Stored in one transaction. Response carries ack_seq	Device deletes the acknowledged samples
Heartbeat with gps_status NO_FIX	Device health ONLINE. Hardware fix marked invalid	Source may switch to the phone

Hardware message	Software action	Result users or staff see
Heartbeat stops during ignition ON	Device DELAYED, then OFFLINE	Technician ticket at OFFLINE
IGNITION_OFF during a shift	Starts the idle timer for auto-end	Driver is prompted after 20 minutes without movement
SOS event	Creates an incident, or joins an open one for the same bus. Starts tier 0 alerts at once. Returns incident_id	Recipients alerted within 5 seconds. Device LED confirms
SOS with triggered_at more than 60 seconds old	Incident marked DEVICE_DELAYED	Coordinator sees the delay
LOW_VEHICLE_BATTERY	Opens a ticket. Notifies technician and coordinator	Ticket on the technician's list
MODULE_ERROR, three times in 24 hours	Opens a ticket	Ticket on the technician's list
FIRMWARE_FAILED	Opens a ticket. Pauses the rollout for this device	Technician sees the paused rollout
Request refused with 401 or 403	Records an authentication failure. Device stops	Device health shows OFFLINE once heartbeats stop
Unknown event type	Stored as IGNORED	None
CONFIG_APPLIED	Logged	None

12. Failure modes and behaviour

Each row shows what the device does, what the software does, and what alert results. Software alerts are defined in software section 22.

Fault	How it is detected	Device action	Software action	Alert
No GNSS fix	No valid fix for 30 seconds	GPS_LOST sent. Keeps searching. Green slow blink	Hardware fix invalid after 60 seconds. Phone takes over in AUTO mode	Ticket if the fault lasts more than 10 minutes
GNSS receiver silent	No output for 10 seconds	Restarts the receiver. MODULE_ERROR after three restarts	Hardware fix invalid	Ticket
No cellular service	Not registered for 60 seconds	Buffers. Retries registration	Hardware stale after 60 seconds	Device offline alert if it lasts during a shift
SIM missing or locked	SIM status in AT responses and heartbeat	Red LED. Buffers. Cannot send	Silence	Technician ticket
Module hang	No AT response for 30 seconds	Restarts the module. Then pulses RESET	Silence until recovery	MODULE_ERROR
Controller hang	Hardware watchdog expires	Reset. BOOT sent with reason WATCHDOG	Gap in data	Ticket after three in 24 hours
Power dip or brownout	Supply below threshold at reset	Reset. BOOT sent with reason BROWNOUT. Buffer kept in flash	Gap in data	Ticket after three in 24 hours
Vehicle battery low	ADC below the class threshold for 30 seconds	LOW_VEHICLE_BATTERY sent. Stops transmitting at cut-off	Ticket	Coordinator and technician
Backup battery low	ADC below its threshold	LOW_DEVICE_BATTERY sent	Ticket	Technician
SOS button stuck	Pressed for more than 10 seconds	Sends one SOS. Ignores the button until released	One incident	Technician notified through the incident
Token rejected	401 or 403 from the server	Stops. Red LED. Rechecks every 15 minutes	Authentication failure recorded	Security review if the count is high
Server unavailable	5xx or timeout	Backs off with jitter. Buffers	Gap in data	Operational alert if long

Fault	How it is detected	Device action	Software action	Alert
Clock wrong	No GNSS time, or skew reported	Sends CLOCK_SYNC when corrected	Clock-skew alert	Technician
Buffer full	Capacity reached	Drops the oldest records. Sends BUFFER_OVERFLOW	Gap recorded	Ticket if repeated
GNSS antenna disconnected	No satellites for a long time	GPS_LOST sent	Hardware fix invalid	Ticket
Frozen GNSS (position fixed while moving)	Cannot be seen by the device	No action	Frozen-position alert after 5 minutes	Technician
Tamper (only if fitted)	Switch opens	TAMPER sent. Red LED	Ticket	Coordinator and technician
Overheating (only if a sensor is fitted)	Temperature above the limit in heartbeat	Flag in heartbeat	Ticket if sustained	Technician

13. Device identity, provisioning, and tokens

13.1 Identity

- device_id: HW_ followed by 3 or 4 digits. Printed on the enclosure label, and encoded in a provisioning QR code.
- Recorded at provisioning: serial number, IMEI of the module, and ICCID of the SIM.
- A device is never re-used under a new device_id. Replacement creates a new device_id.

13.2 Provisioning steps (technician)

1. Create the device records as a batch in the console (software section 16.5). Status becomes PROVISIONED.
2. The server generates a device token and shows it once, as text and as a QR code.
3. Flash the device_id and the token to the controller over the serial port, using the provisioning tool, in production mode.

4. Set the vehicle voltage class and the default configuration.
5. Power the device and confirm that its first authenticated request succeeds. The server then changes the status to ACTIVE.
6. Print and fix the label. Record the step in the provisioning log.

13.3 Token storage

- The token is stored in protected flash, with read protection enabled where the controller supports it.
- Limit recorded: if the controller's flash can be read with physical access, the token can be extracted. The control for this case is revocation at the server (software section 11.1). Per-device tokens mean one extracted token affects one bus only.
- The token is never printed to the serial console, written to logs, or stored in the source repository.

13.4 Rotation

Follow software section 11.1. The device requests a new token using its current token, stores the new token, and uses it. The old token is retired on the first successful use of the new one.

13.5 Replacement and spares

Spares are provisioned before they are stored, so they can be fitted without delay. A replaced device is revoked or retired at the server, and its history stays attached to it.

13.6 Prohibited practices

- Shared tokens across devices.
- Tokens in the code repository, build images, or configuration files.
- Tokens in logs, serial output, or console screens in production.
- Skipping TLS certificate verification in production.

14. Installation procedure

1. Confirm the bus voltage class (12 V or 24 V). The device is set to match at provisioning. Do not change it on site without a record.
2. Choose the enclosure location (section 7). Mark the mounting points.

3. Fit the fuse holder on the battery-side feed, as close to the battery as the design states.
4. Connect power and the single-point ground. Use the wire gauge set by the design.
5. Connect the ignition sense to the ignition-switched circuit identified in the bus's electrical diagram. Confirm the circuit with a multimeter before connecting.
6. Mount the LTE antenna. Route its cable at least 10 cm from high-current harnesses. Tighten the connector to the datasheet torque.
7. Mount the GNSS antenna with a clear view of the sky.
8. Insert the IoT SIM. Confirm the APN and that the PIN is disabled.
9. Mount the SOS button within the driver's reach, not blocked by the steering wheel or seat. Connect it and label it.
10. Mount the status LED and buzzer where the driver and faculty can see and hear them.
11. Fix the enclosure on vibration mounts. Seal the cable entries.
12. Power on. Confirm that BOOT appears and that a heartbeat arrives within 2 minutes. Confirm a valid GNSS fix within 5 minutes outdoors.
13. Confirm with the coordinator that the device is linked to the correct bus. Check the bus number on the label before the link is made.
14. Run the commissioning checklist (section 15).
15. Record the installation: date, bus, device_id, technician, and any photographs. Photographs are kept in the installation log, not in the apps, in the MVP.
16. Sign off in the console.

15. Commissioning checklist

#	Check	Pass when
1	Voltage at the device, at idle and during transmission, with the engine running	Stays above the cut-off with a margin set in the design
2	Fuse and polarity	Fuse fitted to the design value. Polarity correct
3	Ignition sense	IGNITION_ON when the engine starts, and IGNITION_OFF when it stops

#	Check	Pass when
4	Network	Heartbeat shows network REGISTERED and sim READY. Signal better than the design threshold (proposed: minus 95 dBm). CONFIRM
5	GNSS, outdoors	gps_status FIX_VALID within 5 minutes, with 6 or more satellites and HDOP of 2.0 or less (proposed)
6	Samples	ack_seq advances. Valid samples are accepted
7	Drive test, 10 minutes	Samples every 10 seconds while moving. No gap longer than 30 seconds. Bus shows Live, bus tracker
8	SOS drill	Incident arrives marked as a drill. LED and buzzer respond. Coordinator cancels the drill
9	Power interruption, brief	BOOT with reason BROWNOUT on restart. Buffer still holds its samples
10	Network interruption, 2 minutes	Samples are sent in order after reconnection
11	Enclosure and wiring	No bare wire. Cables tied. Glands sealed. Label visible
12	Records	Device linked to the correct bus. Installation record complete

16. Testing and acceptance

16.1 Bench tests

- Power input across the design range, and regulator ripple at the module supply.
- Module AT command set, and the HTTPS handshake against the production certificate chain (software section 34.7).
- GNSS time to first fix, cold and warm.

- Buffer write, read, and overflow behaviour.
- Watchdog reset of the controller, and module restart and RESET.
- Button debounce and stuck-button rule.
- LED states and buzzer loudness.
- Ignition threshold, including transients.
- Firmware update with power removed in the middle of the write. The unit must roll back cleanly.

16.2 Vehicle tests

- Vibration to the agreed profile.
- Heat soak and cold start.
- Cranking dips and electrical transients, checked in the lab against the vehicle specification.
- Two-hour soak at idle, and in motion.

16.3 Failure injection

Antenna disconnection (LTE and GNSS). SIM removal. Blocked network route to the server. Token revocation. Power cut. Module hang. Server 503 storm. Clock set two hours wrong. Each case has an expected device behaviour from section 12.

16.4 Release acceptance

- All bench tests pass.
- Vehicle tests pass.
- Two buses run for two weeks with zero Severity 1 defects.
- Contract tests in software section 32 pass for the firmware build.
- The battery drain test passes (section 6.2).

16.5 Pilot

Three buses for two weeks. Then a 30-day soak on all buses before full rollout.

17. Firmware updates

- Image: signed with the release key. The signature and the SHA-256 are verified before installation (software section 33.8).

- Slots: the new image is written to the inactive slot. After reboot, the device confirms the new version by sending a heartbeat with the new fw_version within 5 minutes. If it does not, it rolls back on the next boot. The ESP8266 flash layout must support two slots, or the team uses a bootloader that keeps a backup image. CONFIRM.
- Conditions: updates run only while the bus is parked with ignition OFF, and the vehicle voltage is above the update threshold (proposed: 12.0 V, or 24.0 V for the 24 V class). CONFIRM.
- No update starts while an SOS is pending.
- Rollout: canary of 5 buses, then 25 percent, then the full fleet, with 48 hours between steps (software section 33.8).
- Versions: fw_version is sent in every heartbeat and every batch.
- Module firmware: the EC200U-CN firmware version is recorded at provisioning. Module updates are tested on the bench first, then rolled out in the same way as firmware updates.

18. Security and tamper

- Production build: serial console disabled, debug output off, and no secrets printed.
- Programming access: locked. The enclosure cover uses tamper-evident screws or seals.
- Tamper switch: optional (section 5.10). CONFIRM whether it is fitted.
- SIM: IoT plan with PIN disabled. SIM changes are logged.
- Theft: revoking the device at the server stops its data. The coordinator is alerted.
- Flash protection and secure boot: enable where the chosen controller supports them. CONFIRM the feasibility on the ESP8266. Any limit found is recorded.
- TLS: certificate verification is required in production. The trust store is set in the EC200U-CN configuration and tested against the production chain.
- Time before TLS: TLS fails if the clock is wrong. The device gets a valid time from GNSS or the network before its first HTTPS request.

19. Compliance and certification

- Radio approvals: the EC200U-CN module and the finished device must hold the Indian approvals that apply to each. The supplier confirms the module's approvals. The college's legal team and the supplier confirm what the finished device needs before it

is used on public roads. Approvals may come from bodies such as the Wireless Planning and Coordination wing or the Telecommunication Engineering Centre, as applicable. CONFIRM.

- Electrical and EMC: the design follows the supplier's guidance for vehicle electrical environments. It is checked on the bench and in the vehicle.
- Materials: RoHS statements from suppliers are kept in the project file.
- Battery: the backup pack's safety and transport documents are kept in the project file. The pack is mounted in a fire-resistant location.
- Data: device data goes only to the college's server (software section 29).

20. Maintenance and spares

- Preventive maintenance every 6 months: connectors, antennas, SIM, enclosure seals, fuse, wiring, and battery health if fitted. Results are logged.
- Corrective replacement: follow the replacement wizard (software section 16.5). Retire the old device, fit a pre-provisioned spare, and link it.
- Spares: 5 percent of the fleet, and never fewer than 2 units. Stored in a dry, temperature-controlled place, with batteries at the supplier's storage charge.
- Fault triage: check the console first (heartbeat, GNSS status, SIM status). Then visit the bus and check the fuse, SIM, antenna connector, and power.
- Repairs: units are returned to the bench. A repaired unit is re-provisioned before reuse.
- Records: installation log, maintenance log, and replacement log.

21. Responsibilities and handover

R: responsible. A: accountable. C: consulted. I: informed.

Activity	ECE team	Software team	Technician	Coordinator	Installer
Hardware design and bill of materials	R, A	C	I	I	I
Firmware	R, A	C (contract)	I	I	I
Server contract (software sections 11 and 12)	C	R, A	I	I	I
Device provisioning	C	I	R, A	I	I

Activity	ECE team	Software team	Technician	Coordinator	Installer
Token issue and rotation	I	R	A	C	I
Linking a device to a bus	I	I	C	R, A	I
Physical installation	C	I	A	I	R
Commissioning sign-off	C	I	R	A	I
Firmware rollout	R (build)	C	R (rollout)	A	I
Maintenance and spares	C	I	R, A	I	I
Device fault response	C	C	R	I	I
Compliance approvals	R (device)	I	I	A	I

Handover to the ECE team, at the end of the MVP build, includes: schematics, bill of materials, firmware source without secrets, the provisioning tool, bench test scripts, installation drawings, the maintenance log template, datasheets, and approval certificates.

22. Open questions for the ECE team (CONFIRM)

1. Controller: ESP8266 or ESP32? The pin count in section 5.3 favours the ESP32.
2. EC200U-CN order code: does it include GNSS? If not, which receive-capable serial port reads the NEO-6M class receiver?
3. Vehicle voltage classes across the fleet: 12 V, 24 V, or both, and the count of each.
4. Backup battery: include or not, runtime, chemistry, and supplier.
5. LTE antenna: type, location, and maximum cable loss.
6. GNSS antenna: type, location, and bias supply.
7. Door sensor: fitted or not, and which pin.
8. Tamper switch: fitted or not, and which pin.
9. RFID reader: confirm phase 2, and the card type for the college.
10. Enclosure: rating, operating temperature range, and mounting location.
11. Installation: who installs, and how long each bus takes.
12. Approvals: which Indian approvals apply to the module and to the finished device, and who obtains them.
13. Firmware slots and update path on the chosen controller.

14. SIM plan: data cap, APN, and vendor.
15. Vehicle cut-off voltages, to confirm sections 6.3 and 17.
16. Logic levels: confirm the EC200U-CN serial voltage in its datasheet, and the level shifter design.

23. Glossary

Term	Meaning
AT command	Text command used to control the cellular module
APN	Access Point Name, the network setting for mobile data
boot_id	Random identifier generated at each boot of the device
Buffer	Flash storage that holds samples until the server acknowledges them
DELAYED, OFFLINE	Device health states defined in software section 11.6
GNSS	Global navigation satellite system receiver, used for position and time
HDOP	Horizontal dilution of precision. Lower is better
IP54	Enclosure rating for dust and splashing water
Load dump	A short, high-voltage spike when a battery is disconnected while charging
Opto-isolator	Component that passes a signal across an electrical barrier
Level shifter	Circuit that converts between two logic voltages, such as 1.8 V and 3.3 V
LTE	Mobile data standard used by the module
NMEA	Text format used by GNSS receivers
Watchdog	Timer that resets a device that stops responding

Term	Meaning
ack_seq	Highest sequence number the server has accepted. The device deletes samples up to it
device_id	Unique identifier of a tracker, in the form HW_nnn
TLS	Encryption protocol used for HTTPS
TVS diode	Protection component that absorbs voltage spikes
UART	Serial communication interface